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Postdisaster image-based damage detection and repair cost estimation of reinforced concrete buildings using dual convolutional neural networks

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Abstract

Reinforced concrete (RC) buildings are commonly used around the world. With recent earthquakes worldwide, rapid structural damage inspection and repair cost evaluation are crucial for building owners and policy makers to make informed risk management decisions. To improve the efficiency of such inspection, advanced computer vision techniques based on convolutional neural networks have been adopted in recent research to rapidly quantify the damage state (DS) of structures. In this article, an advanced object detection neural network, named YOLOv2, is implemented, which achieves 98.2% and 84.5% average precision in training and testing, respectively. The proposed YOLOv2 is used in combination with the classification neural network, which improves the identification accuracy for critical DS of RC structures by 7.5%. The improved classification procedures allow engineers to rapidly and more accurately quantify the DSs of the structure, and also localize the critical damage features. The identified DS can then be integrated with the state-of-the-art performance evaluation framework to quantify the financial losses of critical RC buildings. The results can be used by the building owners and decision makers to make informed risk management decisions immediately after the strong earthquake shaking. Hence, resources can be allocated rapidly to improve the resiliency of the community.

1 | INTRODUCTION

Reinforced concrete (RC) buildings are the most prevalent structural systems constructed worldwide. With many of these buildings built in high seismic zones, the performance of an RC building after strong earthquake shaking is becoming a significant concern for many building owners. When an earthquake happens, decision makers such as city planners and emergency management departments need first-hand response to allocate resources to manage the damaged infrastructure. This requires rapid performance assessments of the facilities. Traditional postearthquake inspections are performed manually and the results may be relatively coarse and highly reliant on the proper training of the inspectors and qualitative engineering judgments. The processing time may also be very long, due to a large amount of data processing required. These deficiencies can be overcome if the current manual evaluation processes are fully automated. Research on automation of inspection practices using computer vision-based methods (Hoskere, Park, Yoon, & Spencer, 2019; Jahanshahi, Kelly, Masri, & Sukhatme, 2009; Koch, Georgieva, Kasireddy, Akinci, & Fieguth, 2015; Yeum, & Dyke, 2015) has greatly advanced in recent years. In the past, computer vision-based methods were developed based

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on conventional image processing techniques (IPTs). However, these methods are relatively time consuming and not robust against background noises. Hence, it is ineffective to apply in practice.

Significant achievements have been made in recent years in computer vision with the advancement of artificial neural networks. The development of artificial neural network can be generally divided into three phases. The first phase can be dated back to the 1940s–1960s, where the theories of biological learning (McCulloch & Pitts, 1943) and the first artificial neural network such as Perceptron (Rosenblatt, 1958) were implemented. The second period happened during the 1980–1995 period, where back-propagation technique (Rumelhart, Hinton, & Williams, 1986) was developed to train a neural network with one or two hidden layers. During the 1990s, artificial neural network evolved to deep neural networks (DNNs), where multiple layers can be trained through back-propagation algorithm. One such application was the work done by LeCun, Bottou, Bengio, and Haffner (1998) for document recognition. The third phase of neural networks (also named deep learning) began with the breakthrough in 2006 when Hinton, Osindero, and Teh (2006) demonstrated that a so-called deep belief network could be efficiently trained using greedy layer-wise pretraining strategy. With the fast growing and optimization of the deep learning algorithms, the increasing size of training data, as well as enhanced computational power, convolutional neural network (CNN, or ConvNet), which is a class of DNNs, has been advancing rapidly. Unlike traditional neural networks that utilize multiple fully connected (FC) layers, the hidden layers of a CNN typically include a series of convolutional layers that convolve with a multiplication or other dot product through learnable filters. In recent years, CNNs have dominated the fields of computer vision, speech recognition, and natural language processing.

Within the field of computer vision, CNN such as the AlexNet, developed by Krizhevsky, Sutskever, and Hinton (2012), has shown a substantial increase in accuracy and efficiency—more than any other algorithms. With the success of AlexNet, CNNs have been successfully applied in computer vision for classification, object detection, semantic segmentation, and visual object tracking. In addition to AlexNet, other deeper CNNs such as VGG Net (Simonyan & Zisserman, 2014), Google Net (Szegedy et al., 2015), Deep Residual Net (He, Zhang, Ren, & Sun, 2016), DenseNet (Huang, Liu, Van Der Maaten, & Weinberger, 2017), and MobileNet (Sandler, Howard, Zhu, Zhmoginov, & Chen, 2018) have been developed.

CNNs have been successfully applied in civil engineering applications for image classifications. These include metal surface defects detection (Soukup & Huber-Mörk, 2014), postdisaster collapse classification (Yeum, Dyke, Ramirez, & Benes, 2016), joint damage detection through a one-dimensional CNN (Abdeljaber, Avci, Kiranyaz, Gabbouj, & Inman, 2017), concrete crack detection using a sliding window technique (Cha, Choi, & Büyüköztürk, 2017), pavement crack detection (Vetrivel, Gerke, Kerle, Nex, & Vosselman, 2018; Zhang et al., 2017), structural damage detection with feature extracted from low-level sensor data (Lin, Nie, & Ma, 2017), and structural damage classification with the proposal of Structural ImageNet (Gao & Mosalam, 2018). Apart from the CNN-based classification, other powerful classification algorithms such as enhanced probabilistic neural network (EPNN) with local decision circles and the new neural dynamic classification (NDC) were successfully developed in recent years (Ahmadlou & Adeli, 2010; Rafiei & Adeli, 2017b). The noticeable applications of such recent algorithms in civil engineering can be found in damage detection in high-rise buildings using neural dynamics classification (Rafiei & Adeli, 2017c), development of earthquake early warning system (Rafiei & Adeli, 2017a), structural reliability analysis (Dai, 2017), estimation of concrete properties (Rafiei, Khushefati, Demirboga, & Adeli, 2017), the global and local assessment of structural health condition using unsupervised deep Boltzmann machine (Rafiei & Adeli, 2018b), and construction cost estimation (Rafiei & Adeli, 2018a).

In addition to classification, CNNs can be used in the field of object detection, which involves classification and localization of an object. Prior to the use of CNNs, object detection was dominated by the use of histogram of oriented gradients (HOG; Dalal, & Triggs, 2005) and scale-invariant feature transform (SIFT; Lowe, 2004). In 2014, Girshick, Donahue, Darrell, and Malik (2014) proposed the region-based CNNs (R-CNNs), which utilizes the region proposal function (RPF) to localize and segment objects. It significantly improved the global performance compared to the previous best result on PASCAL visual object classes (VOC) challenge 2012. The PASCAL VOC challenge ran each year from 2005 to 2012, which provides a benchmark in visual object category recognition and detection with a standard data set of images and annotation, and standard evaluation procedures. Discussion of object detection proposal methods can be found in Hosang, Benenson, and Schiele (2014) and Hosang, Benenson, Dollár, and Schiele (2015). Further, the Fast R-CNN (Girshick, 2015) and the Faster R-CNN (Ren, He, Girshick, & Sun, 2017) were developed to improve the speed and accuracy of the R-CNN. Region-based CNN methods (e.g., RCNN, Fast-RCNN, and Faster-RCNN) have been successfully implemented in civil engineering applications. Cha, Choi, Suh, Mahmoudkhani, and Büyüköztürk (2018) used the Faster-RCNN to detect multiple structural damage types such as steel delamination, steel corrosion, bolt corrosion, and concrete cracks. Xue and Li (2018) demonstrated the efficiency of Faster R-CNN for shield tunnel lining defects detection compared to traditional inspection methods. Li, Yuan, Zhang, and Yuan (2018) achieved near-real-time concrete defect detection with geolocalization using a unified vision-based

methodology. Liang (2019) applied deep learning with Bayesian optimization for RC bridge damage detection.

Although many previous studies can provide reasonable accuracy, they are still relatively slow in terms of achieving real-time practical application when the images were recorded with high frame per second (FPS). To address this deficiency, Redmon, Divvala, Girshick, and Farhadi (2016) presented YOLO (i.e., You Only Look Once) for real-time object detection. Although YOLO is extremely fast, it makes more localization errors and achieves relatively low recall compared to region-based CNN methods. To further improve recall and localization accuracy, Redmon and Farhadi (2017) developed the YOLOv2 algorithm. They have shown that YOLOv2 significantly improves the recall and localization accuracy while still maintaining the speed to be 10 times faster in FPS compared to Faster-RCNN on the VOC 2007 database.

CNN-based classification for civil engineering applications has been hampered by limited training data (Gao & Mosalam, 2018). In general, a single classification model can provide reasonable accuracy if the training data cover a wide range of hidden features. However, even when the size of the training data is sufficiently large, the classification model may still not perform well if the training data are not properly preprocessed to identify the localized damage (Gao & Mosalam, 2018). For example, an image may contain multiple damage states (DSs) where a portion of the structure has fractured, whereas the other part of the structure remains undamaged.

Moreover, although region-based CNN methods have been widely applied in civil engineering, there remains an almost little to no attempt to use the regression-based detection methods such as YOLOv2, for structural damage detection. Therefore, the main contributions of this article are as follows: (1) established component training data that follow the codified DS classification of the RC columns; (2) successfully developed and applied YOLOv2 object detection network to identify the critical damage feature of RC columns; (3) proposed and successfully implemented the dual CNN methods that incorporate the classification network and YOLOv2 object detection network to improve the accuracy achieved by a single classification network; (4) introduced performance-based assessment framework to quantify financial losses, which can be used by the decision makers for rapid emergency management and resources allocation, thus improving the regional seismic resiliency of the city.

2 | METHODOLOGY

2.1 | Rapid performance assessment framework

Figure 1 shows the framework proposed in this study to quantify the financial loss of the RC buildings after strong earthquake shaking. First, system-level and component-level images for a single building are collected, which can be achieved by unmanned aerial vehicles (UAVs; Ham, Han, Lin, & Golparvar-Fard, 2016). In an ideal situation, images of the systems and components should be taken from multiple views and the most severe damage status should be considered to facilitate the comprehensive evaluation. The system-level images are assessed by a system-level classification network to confirm if the building has collapsed. If the system-level collapse is identified, the replacement cost of the building should be used. If the building is identified as noncollapse, the component-level images are fed into component-level classification and detection networks. Once the component DSs are identified, the corresponding repair costs for the components are identified from the ATC-58 (2007) fragility database. Finally, the total repair costs of the building are summed up by adding the total repair quantities from all structural and nonstructural components taking into account their suitable unit cost distribution. This fragility database is one of the essential products of ATC-58 project established by the Applied Technology Council (ATC) in contract with the Federal Emergency Management Agency (FEMA) to develop FEMA P-58 Seismic Performance Assessment of Buildings, Methodology, and Implementation (also known as Performance-Based Earthquake Engineering [PBEE]). Implementations of the PBEE framework for cost evaluation of buildings have been widely attempted (Goulet et al., 2007; Mitrani-Resier, Wu, & Beck, 2016; Yang, Moehle, Stojadinovic, & Der Kiureghian, 2009). Although this study focuses on the development of dual CNN methods that employs the state-of-the-art YOLOv2 object detection algorithms to more accurately classify the structural DS, it also makes the first attempt to integrate this fragility database with the proposed deep learning methods to facilitate cost evaluation of RC structures.

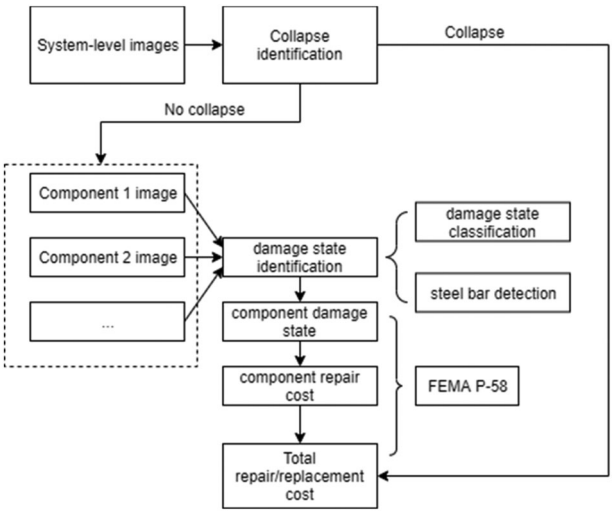


FIGURE 1 Flowchart of cost evaluation scheme

2.2 | CNN classification

In this research, CNNs were used to identify the DSs of the building and components. Typical CNN involves multiple types of layers, including convolution (Conv) layers, rectified linear unit (ReLU) layers, pooling layers, FC layers, and loss layer (e.g., Softmax layer). The Conv layer combined with the subsequent layer, ReLU, constitute the essential computational block for CNNs. This is the feature that distinguishes CNNs from the traditional FC deep learning network. One of the advantages of CNNs is that it drastically improves the computational efficiency from the traditional neural network because the number of training parameters enclosed in the filter of CNNs is significantly less than the number of weights utilized by FC layers, which are the only layers presented in the traditional feed forward neural network. Besides, CNNs preserve the spatial locality of pixel dependencies and enforce the learnable filters to achieve the strongest response to a local input pattern.

During the forward pass, the output from the previous layer is convolved with each one of the learnable filters, which yields a stack of two-dimensional arrays. Applying the desired nonlinear activation function (such as ReLU) to these two-dimensional arrays leads to a volume of two-dimensional activation maps. After a single or multiple Conv–ReLU blocks, the pooling layer is introduced, which is a form of nonlinear down-sampling. The objective of the pooling layer is to reduce the number of parameters to improve the computation efficiency. During the pooling process, the input image is partitioned into subregions, which may or may not overlap with each other. If max pooling is used, the maximum value of each subregion is taken.

Following several Conv–ReLU blocks and pooling layers, the resulting layer is transposed to an FC layer. The output can be computed as matrix multiplication followed by a bias offset, which then substitutes into activation function. For example, in VGG-19, the CNNs end up with 3 FC layers with the dimension of 4096, 4096, and 1000, respectively. In addition, due to the fact that FC layers occupy most of the parameters in the entire CNN, they are prone to overfitting, which can be alleviated by incorporating dropout layers. The idea of using dropout layers is to randomly remove FC layers, which can improve the computation efficiency and has proven to alleviate the concern for overfitting. In VGG-19, 50% dropout is applied to the FC layers. Finally, the output of the last FC layer is passed to a loss layer (i.e., Softmax in this study), which determines the probability of each class (i.e., the confidence of the model to perceive the input image being each class). The result of the classification is recognized for the output with the highest probability for each class.

2.3 | System-level collapse recognition

Based on the CNNs presented, the status of the RC building can be classified as collapse or noncollapse. Multiple pretrained models can be used to facilitate the training process. In this study, transfer learning from the three pretrained models including AlexNet (Krizhevsky et al., 2012), VGG-19 (Simonyan & Zisserman, 2014), and ResNet-50 (He et al., 2016) is applied for the binary classification task. Transfer learning is a new machine learning technique that takes advantage of certain pretrained models in the source domain and fine-tunes part of the parameters with a few labeled data in the target domain, which can greatly promote the training process in the situation of data scarcity.

In deep learning, there is a trend to develop a deeper and deeper network, which aims at solving more complex task and improving the performance. However, research has shown training of DNNs becomes difficult and the accuracy can reach plateau or even degrade (He et al., 2016). ResNets were developed by He et al. (2016) to solve the problems where the shortcut connections were proposed. It has been demonstrated that training this form of networks is easier than training plain deep CNNs and the problem of accuracy deterioration is resolved. The complete architecture of ResNet-50 adopted in this study is shown in Figure 2. The ResNet-50 contains a sequence of convolution–batch normalization–ReLU (Conv-BN-ReLU) blocks. Batch normalization is added right after each convolution and before ReLU activation to stabilize training, speed up convergence, and regularize the model. After a series of Conv-BN-ReLU blocks, global average pooling (GAP) is performed to reduce the dimensionality, which is then followed by the FC layer associated with Softmax function.

Due to the limited number of images for civil engineering applications, 686 images are collected from datacenterhub.org at Purdue University and Google images, of which 240 images are related to the collapse of buildings and 446 images of noncollapsed buildings. The image preprocessing is conducted to reduce the inconsistency in image classification following the same approach adopted by Gao and Mosalam (2018). The preprocessed images will be resized appropriately to 224×224 or 227×227 pixels (depending on what network is chosen) before being fed into the CNNs for state and damage classification. The performance of the model is verified through the training and validation process. In this case, 80% of the collected images are chosen as the training data and the rest are chosen as the testing data. Further, within the training set, 20% of the images are set as the validation data and the remaining images are used to train the model. Therefore, $686 \times 0.8 \times 0.8 \approx 439$, $492 \times 0.8 \times 0.2 \approx 110$, and $492 \times 0.2 \approx 137$ images are allocated for training, validation, and testing purposes, respectively.

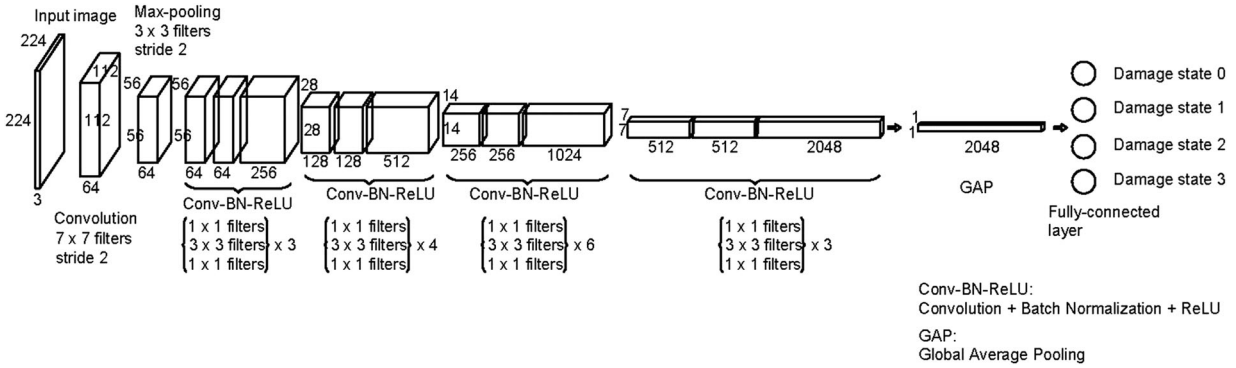


FIGURE 2 Architecture of ResNet-50

TABLE 1 Description of damage state classes

DS index	Description
0	No damage
1	Light damage: visible narrow cracks and/or very limited spalling of concrete
2	Moderate damage: cracks, large area of spalling concrete cover without exposure of steel bars
3	Severe damage: crushing of core concrete, and/or exposed reinforcement buckling or fracture

2.4 | Component-level DS identification

As per the proposed evaluation scheme depicted in Figure 1, if the RC building is identified as noncollapse, the subsequent step is to determine the DS of the structural components. In this study, the definition of several DSs for RC structural columns is introduced as shown in Table 1, which follows the ATC-58 DS statement for RC beam-column joints. Similar evaluation approaches have been shown and practically implemented for many years as demonstrated by Nakano, Maeda, Kuramoto, and Murakami (2004) and Maeda, Matsukawa, and Ito (2014).

The RC columns, the critical gravity supporting components of the RC buildings, are selected to demonstrate the component-level classification, detection, as well as cost evaluation. In this regard, a novel dual CNN-based inspection approach (Figure 3) is proposed to facilitate the process. On one hand, the classification model is trained across all the DSs as defined in Table 1. On the other hand, localization of steel bars is implemented using the YOLOv2 object detection approach. The advantage of the object detection approach is its ability to focus on damage-sensitive features (i.e., exposed reinforcement bars in this case), which distinguish DS 3 from DS 0, DS 1, and DS 2. It is noted that the detection of exposed reinforcement is crucial because most of the tensile stiffness and strength of the RC components are contributed by the reinforcement. Therefore, CNN-based detection is employed to reinforce the identification of DS 3 in case the classification fails to classify it. In fact, from the safety point

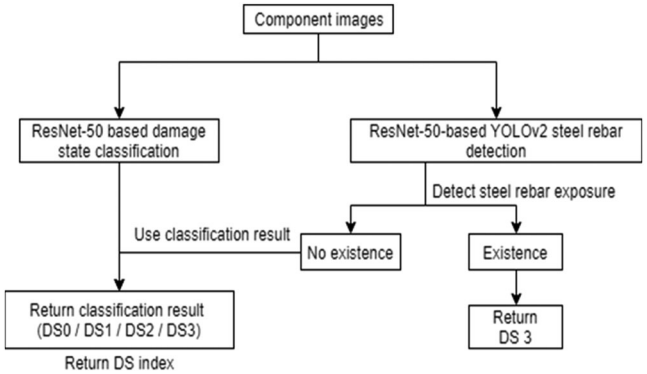


FIGURE 3 Flowchart of component damage state inspection scheme

of view, identification of components in DS 3 condition is critical after the event of earthquake because these components are prone to fail completely in the aftershocks, which may lead to partial or complete collapse of the building and consequently significant increase of repair cost, injuries, and death rate. In other words, it is more conservative to maintain the second object detection network even if in some rare cases, the final DS is identified as DS 3 although the ground truth label is one of the others.

In total, there are 2,260 images collected from the damage survey conducted by Sim, Laughery, Chiou, and Weng (2018), EERI Learning from Earthquake Reconnaissance Archive and Google Image. The number of images for DS 0, DS 1, DS 2, and DS 3 is 496, 404, 580, and 780, respectively. Similar as before, image preprocessing and resizing are applied before training. Also, 80% of the acquired images for each damage class is chosen as its training set and 20% as the testing set. The validation set is chosen as 20% of the training set and the rest of the training set is used to train the model.

2.4.1 | DS classification

As shown in Table 1, four DS classes need to be distinguished. Similar to system-level classification, the pretrained AlexNet, VGG nets, and ResNet-50 are selected for transfer learning.

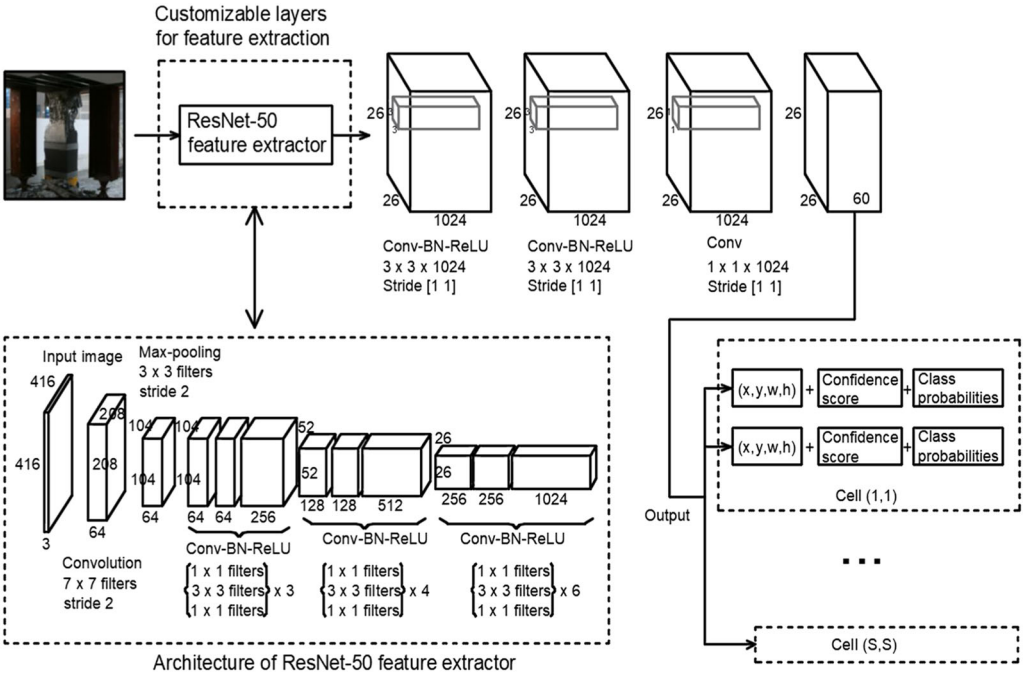


FIGURE 4 The schematic architecture of YOLOv2 built on ResNet-50 for steel reinforcement detection

The trained model with the highest test accuracy is adopted to demonstrate the applicability of the classification of multiple DSs. The construction of the network is similar to the previous one except the last three layers, an FC layer, a Softmax layer, and a classification output layer are updated with new labels and the new number of classes (i.e., four DSs in this case).

2.4.2 | Steel reinforcement object detection

In addition to DS classification, a CNN-based object detection model is also introduced in this study to identify steel reinforcement exposed due to concrete spalling. In comparison to image classification, object detection is one step further which localizes the object within an image and predicts the class label of the object. The output of object detection would be different bounding boxes with their labels in the image. Although R-CNN methods (i.e., R-CNN, Fast R-CNN, Faster R-CNN) have been widely attempted in civil engineering applications, they are still relatively slow for real-time applications. This study designed and applied a specific YOLOv2 object detection network for identification of reinforcement exposure. A detailed comparison between YOLOv2 and other object detection networks is presented in Redmon and Farhadi (2017).

In general, YOLOv2 consists of a customized feature extractor, which is usually a series of Conv-BN-ReLU blocks and pooling layers, followed by localization and classification layers, which predicts the bounding box location and the class score, respectively. In this study, YOLOv2 built on ResNet-50 is adopted for steel reinforcement detection. First, the layers after the third Conv-BN-ReLU block of ResNet-50 (as shown in Figure 4) are removed such that the remaining layers can work as a feature extractor. Second, a detection subnetwork is added, which comprises groups of serially connected Conv-BN-ReLU blocks. Details of layer properties within the detection subnetwork are illustrated in Figure 4. In conclusion, the detection is modeled as a regression problem. The output of the network contains $S \times S$ grid cells of which each predicts B boundary boxes. Each boundary box includes four parameters for the position, one box confidence score (objectness), and C class probabilities. The final prediction is expressed as a tensor with the size of $S \times S \times B \times (4 + 1 + C)$.

The objective of training of neural network is to minimize the multipart loss function as shown in Equation 1 where $I_{i,j}^{obj} = 1$ if the jth boundary box in cell i is responsible for detecting the object, otherwise 0. Similarly, $I_i^{obj} = 1$ if an object appears in cell i, otherwise 0, and $I_{i,j}^{noobj}$ is the complement of $I_{i,j}^{obj}$. The parameters x_i and y_i are the predicted bounding box position, $\hat{x}_i$ and $\hat{y}_i$ refer to the ground truth position, w_i and h_i are the width and height of the predicted bounding box, whereas the associated ground truth is denoted as $\widehat{w_i}$ and $\widehat{h_i}$. The properties of the predicted bounding box are refined based on the predefined anchor boxes as illustrated in Figure 5. The term C is the confidence score and $\hat{C}$ is the intersection over union of the predicted bounding box with the ground truth. The multiplier λ_{coord} is the weight for the loss in the boundary box coordinates and λ_{noobj} is the weight for the loss in the background. As most boxes generated do not contain any objects, indicating the model detects background more frequently than detecting objects, to put more emphasis

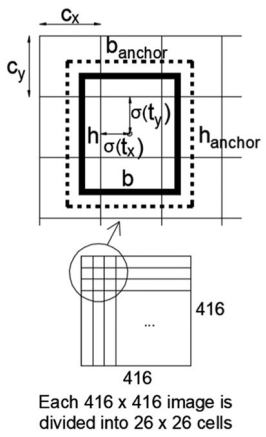


FIGURE 5 Geometric properties of the predicted bounding box and the anchor prior

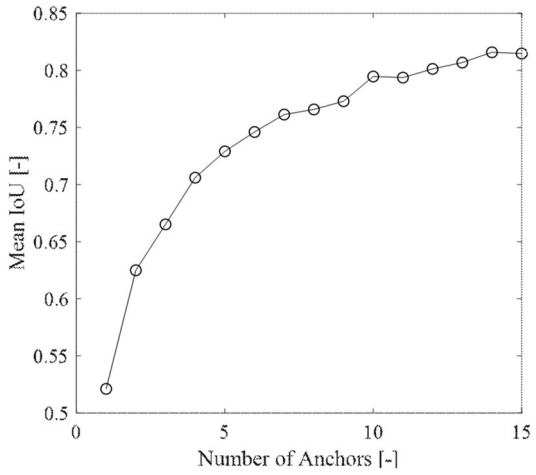


FIGURE 6 Relationship between Mean IoU and number of dimension priors

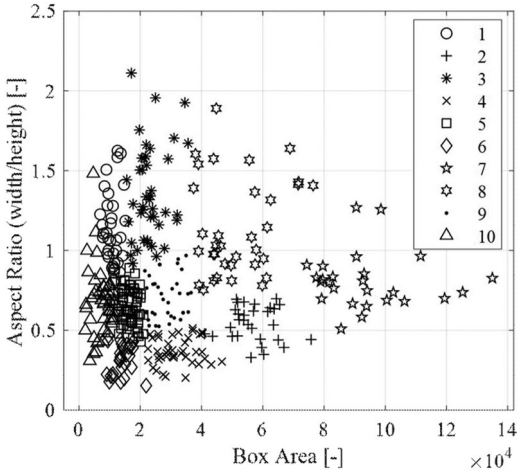


FIGURE 7 Illustration of K-means clustering results with 10 anchor sizes

on the boundary box accuracy, λ_{coord} is set to 5 by default and λ_{noobj} is chosen as 0.5 by default.

$$\lambda_{coord}\sum_{i=0}^{s^2}\sum_{j=0}^{B} I_{i,j}^{obj}\left[\left(x_i-\hat{x}_i\right)^2+\left(y_i-\hat{y}_i\right)^2\right]+\lambda_{coord}\sum_{i=0}^{s^2}\sum_{j=0}^{B} I_{i,j}^{obj}\left[\left(\sqrt{w_i}-\sqrt{\hat{w}_i}\right)^2+\left(\sqrt{h_i}-\sqrt{\hat{h}_i}\right)^2\right]+\sum_{i=0}^{s^2}\sum_{j=0}^{B} I_{i,j}^{obj}\left(C_i-\hat{C}_i\right)^2\lambda_{noobj}+\sum_{i=0}^{s^2}\sum_{j=0}^{B} I_{i,j}^{noobj}\left(C_i-\hat{C}_i\right)^2+\sum_{i=0}^{s^2} I_i^{obj}\sum_{c\in classes}\left(p_i(c)-\hat{p}_i(c)\right)^2 \quad (1)$$

The network learns to adapt predicted boxes appropriately with regards to ground truth data during training. However, it would be much easier for the network to learn if better anchor priors are selected. Therefore, to facilitate the training process, K-means clustering as suggested by Redmon and Farhadi (2017) is implemented to search the tradeoff between the complexity of the model and the number of bounding boxes required to achieve the desired performance. Once the number of anchors is specified, the K-means clustering algorithm takes as input the dimensions of ground truth boxes labelled in the training data, and outputs the desired dimensions of anchor boxes and the mean intersection-over-union (IoU) with the ground truth data. Clearly, the selection of more anchor boxes provides higher mean IoU, but also causes more computational cost. Through the parametric study on the number of anchors, the relationship between the mean IoU and the number of anchors is established in Figure 6, which shows the number of 10 anchors is a reasonable choice, where the mean IoU can reach about 0.8. It should be noted that unlike the original work by Redmon and Farhadi (2017) where the network utilizes five box priors to classify and localize 20 different classes, this study only focuses on the detection of one class (i.e., steel reinforcement), indicating more anchors can be used without losing too much computational efficiency. Figure 7 depicts the dimensional properties of each ground truth box as well as its associated clustering. Anchor dimensions corresponding to each cluster, determined by K-means clustering approach, are reported in Table 2. These anchors will be utilized to determine the bounding box properties as shown in Equations 2–6. In summary, B is chosen as

TABLE 2 Selection of anchor properties

Group	1	2	3	4	5	6	7	8	9	10
Width (pixels)	104	174	174	107	105	67	274	208	138	54
Height (pixels)	98	309	132	285	167	206	338	213	199	77

10. C is equal to 1, which corresponds to steel exposure. As a result, the predicted tensor has the size of $26 \times 26 \times 60$.

The network predicts 10 bounding boxes at each cell in the output feature map. For each bounding box, five coordinates are predicted including t_x, t_y, t_w, t_h, and t_0. As illustrated in Figure 5, the bold solid box is the predicted boundary box and the dotted rectangle is the anchor prior. Assuming the cell is offset from the top left corner of the image by (c_x, c_y) and the anchor box prior has a width of b_{anchor} and height of h_{anchor}, then Equations 2–6 can be derived. Equations 2 and 3 predict the location of the bounding box and Equations 4 and 5 predict the dimensions of the bounding box based on anchor box dimensions. Equation 6 is related to objectness prediction and involves the IoU of the ground truth and the proposed box, and the conditional probability of the class given that there is an object.

$$b_x = \sigma\left(t_x\right) + c_x \tag{2}$$

$$b_y = \sigma\left(t_y\right) + c_y \tag{3}$$

$$b = b_{anchor} e^{t_w} \tag{4}$$

$$h = h_{anchor} e^{t_h} \tag{5}$$

$$\Pr\left(object\right) \times IoU(b, object) = \sigma\left(t_0\right) \tag{6}$$

Similar to other CNN models, the YOLOv2 is trained by back-propagation and stochastic gradient descent (SGD). The learning rate is constant and set to 10^{-4} and mini-batch size is set to 16. The input image size is 416×416, which is identical to what has been adopted by Redmon and Farhadi (2017) for fine-tuning detection subnetwork. The training and testing images of YOLOv2 are taken separately from the DS 3 images, which have been used in training and testing of the DS classification model. Data augmentation such as cropping, flipping, small rotation is applied, such that the augmented images still contain the object that needs to be detected. The training is implemented in MATLAB R2019a on two computers: Alienware Aurora R8 (a Core i7-9700K @ 3.60 GHz, 16 GB DDR4 memory and 8 GB memory GeForce RTX 2070 GPU) and a Lenovo Legion Y740 (a Core i7-8750H @2.20 GHz, 16 GB DDR4 memory and 8 GB memory GeForce RTX 2070 max-q GPU).

2.4.3 | DS determination

Postearthquake cost evaluation of the RC building strongly relies on the classification accuracy of DS. A single classification model generally performs well if trained on the large data set, which covers a wide range of hidden features. Besides, it is required that the image scene is properly preprocessed such that the targeted region (i.e., RC column with/without damage in this study) dominates the entire image. Moreover, the classification model may not perform well if different classes have obvious shared features. In case of multiple irrelevant objects in the background or a column with multiple damage features presented in a single image (i.e., crack feature is shared by DS 1, DS 2, and DS 3; the spalling feature is shared by DS 2 and DS 3), the classification model may fail to identify the DS class correctly. For example, an image of column is shown in Figure 8 that shows a lot of small cracks and small spalling, and at the same time also presents exposure of evident steel bars concentrated in relatively small regions. The classification model fails to identify it as DS 3 (i.e., ground truth label) in our experiments. This is interpretable because in this case, the damage features of DS 1, DS 2, and DS 3 are all included in one single image, leading to the fact that the predicted probability for DS 3 is not the highest. This is the moment when detection of steel bars is needed to reinforce the severe DS identification for DS 3 where the exposed reinforcement needs to be specifically captured.

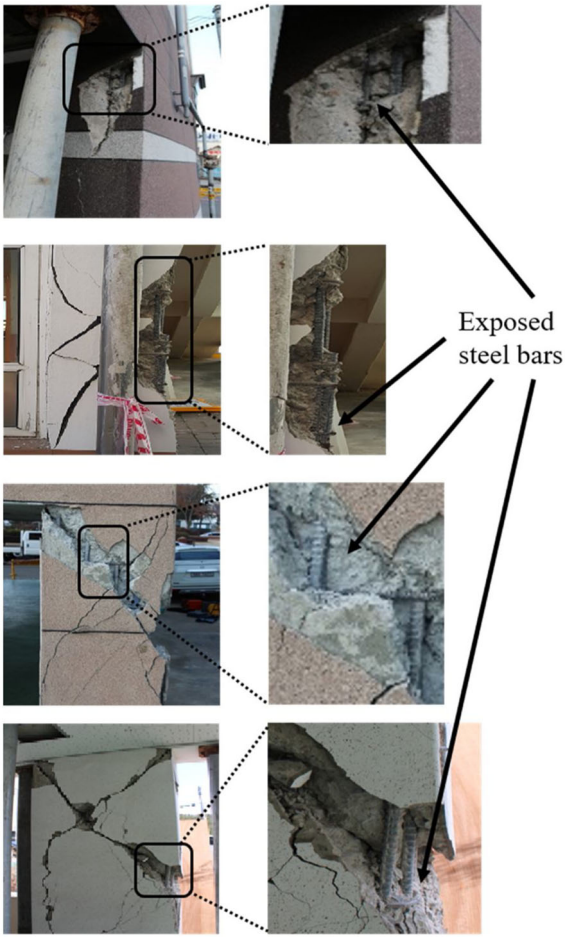


FIGURE 8 Sample images of RC columns that classification model wrongly identifies as DS 2 whereas ground truth is DS 3

In this study, the results of the classification model including the label and its associated probability are obtained.

Meanwhile, the reinforcement detection model checks the existence of exposed steel bars. The final decision on the DS is taking advantage of both outcomes from the classification model and object detection model. As shown in Figure 3, each image is evaluated by the two models in parallel. For a given image, it is first resized to fit the size of input layers of the classification networks and object detection networks, respectively. The classification model predicts the probability of being each DS and takes the one with the highest probability as the result. Meanwhile, the object detection model aims at detecting the exposed steel bars. If the steel bars are not detected, the classification result is directly output as the final inspection outcome. If the steel bars are captured by the detection model, then DS 3 should be returned as the final decision. The proposed dual CNN-based framework builds on the traditional damage classification model, and extends the evaluation scope by analyzing the local details (i.e., steel bars in this case). The object detection model does not change the fundamental diagnosis logic of the classification model but reinforces the identification of DS 3 through localization of exposed bars on top of the classification model. This comes from three bases. First, object detection is a more complex task than classification because it involves both localizing and classifying the target in the scene. Solely relying on object detection results is more likely to lower the overall evaluation accuracy (potentially caused by insufficient recall). Second, there are some other damage features of DS 3 such as crushing of concrete, substantial shear failure mechanism, although the proposed object detection model is trained to detect exposed steel bars only. The identification of such multiple features still partly relies on the classification model. Third, steel reinforcement is one of the most essential load-carrying components in RC columns. The proposed detection network of steel bars introduces one redundancy to reinforce the identification of the most severe damage case, which is more likely to cause an unexpected system failure in response to aftershocks, and consequently higher chances of injuries and death, as well as substantial repair cost and longer repair time.

2.5 | Cost evaluation framework

Once the DS is identified, the repair costs can be calculated using the fragility data presented in the FEMA P-58 project (ATC-58, 2007). Table 3 shows an example of the DS and repair cost for the RC column specified in FEMA P-58 project. The dispersion is defined as the standard deviation of the logarithmic value of the cost (i.e., σ).

FEMA P-58 divided all vulnerable structural components and nonstructural components into fragility groups and performance groups. A fragility group is defined as a group of components that have similarity in construction and installation techniques, modes of damage, probability of inducing damage modes, as well as damage consequences. Besides, a performance group is a subset of fragility group components that are subjected to the same earthquake demands such as story drift or acceleration at a specific floor, in a specific direction.

TABLE 3 Mean value and dispersion of repair cost of reinforced concrete column (e.g., component ID: B1041.031a) at each damage state (DS; ATC-58, 2007)

DS index	Mean cost (US$) at minimum quantity	Mean cost (US$) at maximum quantity	Dispersion
DS 0	0	0	0
DS 1	25,704	20,910	0.39
DS 2	38,978	25,986	0.32
DS 3	47,978	31,986	0.3

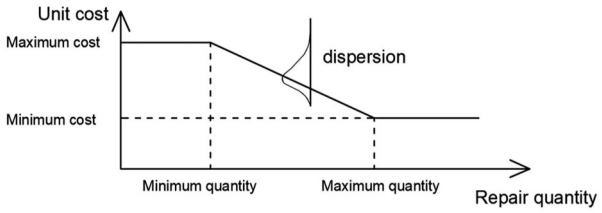


FIGURE 9 Typical consequence function for repair costs

The consequence function for repair costs as defined by FEMA P-58 is shown in Figure 9. The minimum cost refers to the unit cost to conduct a repair action, considering all possible economies of scale (which corresponds to maximum quantity) and operation efficiencies. On the contrary, the maximum cost is the unit cost with no benefits from scale and operation efficiencies, which corresponds to the minimum quantity. If needed, unit repair costs uncertainties can be accounted for using normal or lognormal distribution.

In this study, the cost evaluation scheme is inspired by part of the seismic performance assessment methodology from FEMA P-58. To initiate the proposed cost evaluation scheme, the global images of the building should be assessed by the global classification model to identify whether the building experiences collapse. If collapse occurs, the replacement cost of the building is reported. If the building does not have global failure, the DS inspection approach is continued. In this case, the images of components are processed by the local damage classification model and object detection model in parallel to identify the DS. The cost of the component is retrieved based on the DS identified. Finally, the total repair cost of the building is calculated as the sum of each performance group which contains structural or nonstructural components.

$$\text{Total repair cost} = \sum_{i}^{l}\sum_{j}^{m}\sum_{k}^{n} C_{i,j,k}\left(\sim \log N\left(\mu, \sigma^2\right)\right) \quad (7)$$

TABLE 4 System-level and component-level training parameters and performance of transfer learning from three different pretrained models

Pretrained CNN models	AlexNet	VGG-19	ResNet-50
Input size	227 × 227 × 3	224 × 224 × 3	224 × 224 × 3
Initial learning rate	0.0001	0.0001	0.0001
Regularization factor	0.0001	0.0001	0.0001
Momentum coefficient	0.90	0.90	0.90
System-level testing accuracy	93.15%	95.63%	95.92%
Component-level testing accuracy	85.17%	87.17%	87.47%

where l denotes the number of fragility groups, m denotes the number of performance groups, and n denotes the number of components within each performance group. The term $C_{i,j,k}$ is the repair cost drawn from the normal or lognormal distribution based on the identified DS. Finally, the Monte Carlo process is repeated in accordance with the assumed unit cost distribution for a large number of realizations, of which each represents one total loss value. These realizations are sorted in ascending order and a lognormal distribution is fitted to facilitate the calculation of the probability that total loss will be less than any specific value. The results are presented in a loss curve, which can be used for risk management decisions.

3 | EXPERIMENTS AND RESULTS

3.1 | System-level failure classification

This subsection presents the results from three different pretrained models as shown in Table 4. Although the testing accuracy among all three pretrained models is relatively close, the ResNet-50, which has the deepest architecture, yields slightly higher accuracy than the other two CNN models. The loss and accuracy of ResNet-50 during training are presented in Figure 10. Both the training and validation accuracy exceed 90% after around 80 epochs of training and approaches 100% at the end. It is generally acknowledged that the validation data set can provide an unbiased assessment of a model fit on the training data set (He et al., 2016; Krizhevsky et al., 2012). Increase in the error on the validation data set is a sign of overfitting to the training data set. A high and stable validation accuracy of the validation is observed in Figure 10, which demonstrates the applicability of the system-level classification model for collapse identification. Figure 11 compares the confusion matrices (Kohavi & Provost, 1998) between training and testing results. For instance, 95% of the testing images that have ground truth labels as collapse are successfully predicted whereas only 5% of these images are misclassified as "no collapse." Moreover, sample testing images with probability for its associated class are shown in Figure 12. The trained model can predict the correct class for the images with high probability.

3.2 | Component-level DS identification

3.2.1 | DS classification

Similar to the system-level failure classification, Table 4 presents the identical training parameters and performance comparison of three different pretrained models (AlexNet, VGG-19, ResNet-50) for the classification of the component DSs. In general, all three models have high accuracy, although the ResNet-50 has slightly higher accuracy than AlexNet and VGG-19. The loss and accuracy for ResNet-50 during the training process are presented in Figure 13, which shows both the training and validation accuracy are approaching 100% at the end. The performance of the trained model is confirmed by the confusion matrix for training and testing as shown in Figure 14. Figure 15 shows the classification of a few sample images with correct prediction. The results show that the trained model is able to classify different DSs with reasonably high accuracy, although the classification accuracy with respect to moderate damage (i.e., DS 2) and severe damage (i.e., DS 3) are not as high as that regarding the class of no damage (i.e., DS 0) and light damage (i.e., DS 1). The results reflect there is an increasing difficulty in detecting damage features from DS 0 to DS 3. Basically, there is no damage feature in DS 0 when RC columns are in almost perfect condition in which case the trained CNN model only needs to identify column profile without any extra damage features. Similarly, only cracks and very limited spalling are enclosed in DS 1, where slightly more damage features are introduced compared to DS 0. However, more features are usually observed in DS 2, such as light or severe cracks and a large area of spalling. In the case of DS 3, the model performs reasonably well to detect its own damage features, which include exposure of significant length of steel bars, crushing of concrete, and buckling or fracture of reinforcement. However, it occasionally misclassifies the DS as DS 2, whereas the ground truth is DS 3 (Figure 16). There are two potential reasons. First, DS 2 and DS 3 have many common damage features, such as cracks and a large amount of concrete spalling. Second, exposure of steel reinforcement is not evident in some cases although cracks or significant spalling may be dominant in the entire image (Figure 8). To overcome such deficiency, a novel object detection technique is implemented and combined with the

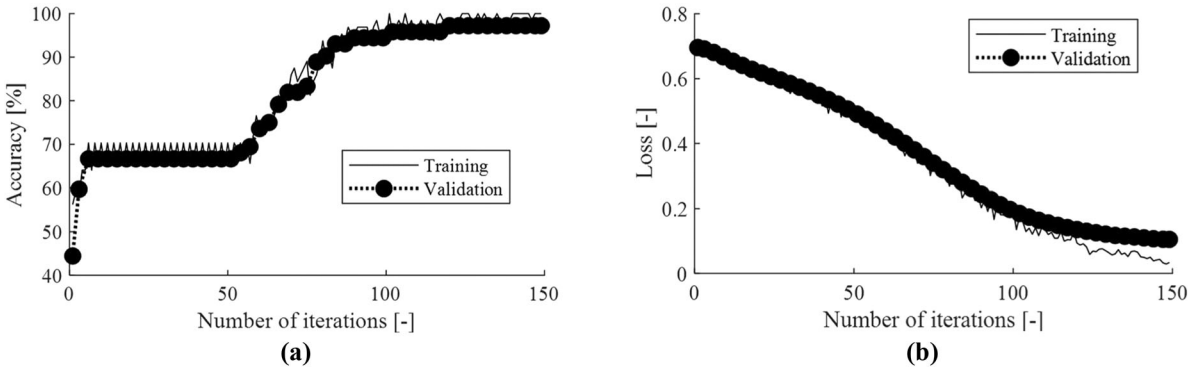


FIGURE 10 System-level collapse identification for training and validation sets using ResNet-50: (a) accuracy and (b) loss

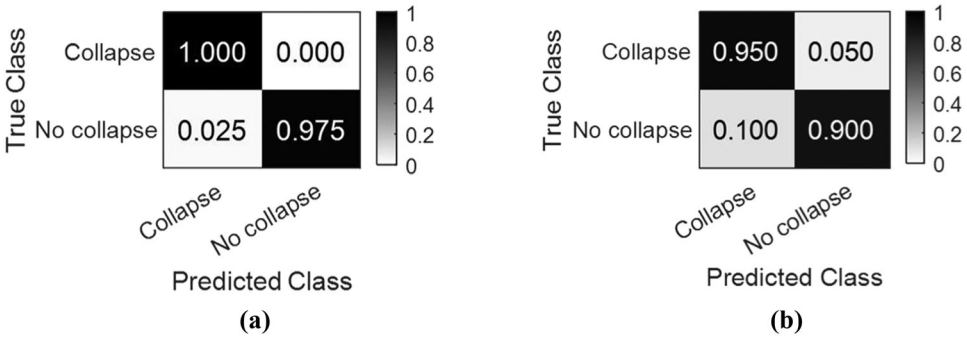


FIGURE 11 System-level collapse versus no collapse: confusion matrices of (a) training set and (b) testing set

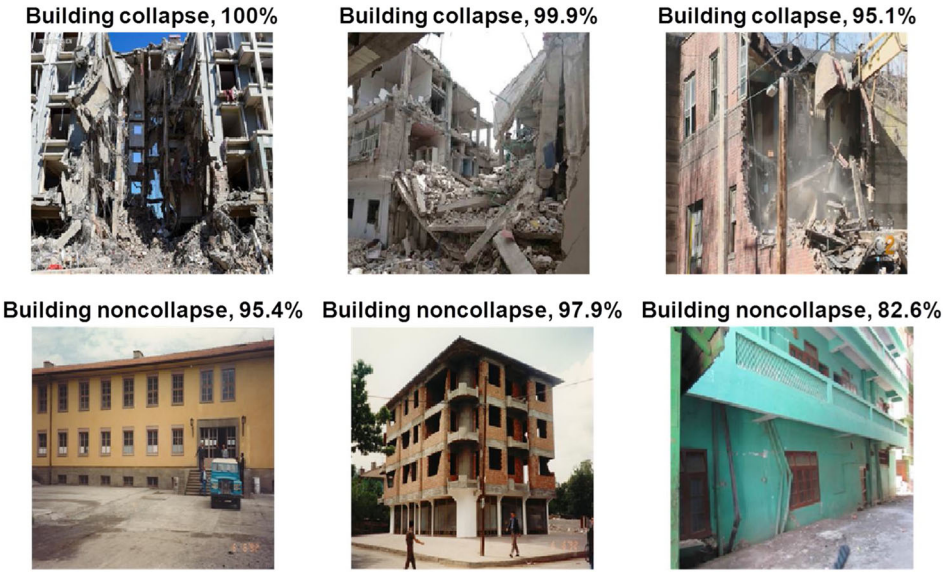


FIGURE 12 Sample testing images of the building with predicted probability for each class

classification technique to identify the DSs (Figure 3). Details of the integrated method are described in Session 3.2.3.

3.2.2 | Steel reinforcement object detection

This subsection presents the results regarding the detection of exposed longitudinal reinforcement in RC columns to demonstrate the applicability of YOLOv2 in this scenario. The performance of an object detector is usually evaluated using a precision–recall curve (Everingham, Van Gool, Williams, Winn, & Zisserman, 2010). A low false positive rate leads to high precision and low false negative rate results in a high recall. In other words, a large area under the recall–precision curve indicates the high performance of the detector with

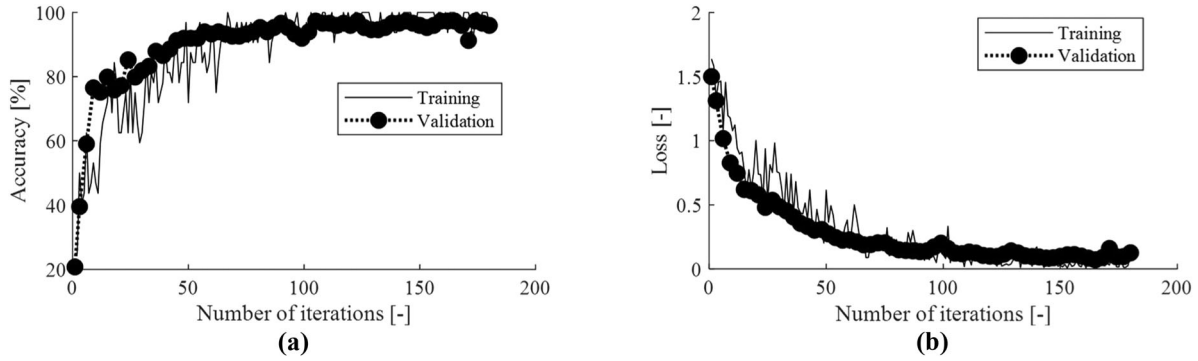


FIGURE 13 Component-level DS classification for training and validation sets using ResNet-50: (a) accuracy and (b) loss

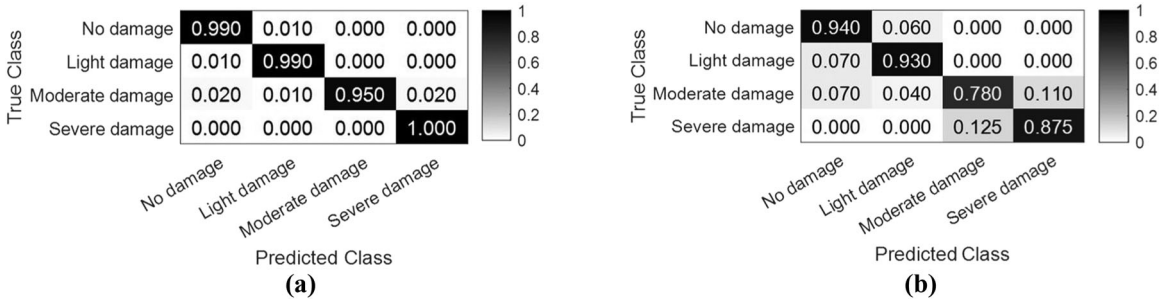


FIGURE 14 Component-level damage state identification: confusion matrices of (a) training (left) and (b) testing set



FIGURE 15 Reasonable prediction of sample testing images of the building with predicted probability for each class



FIGURE 16 Unreasonable prediction of sample testing images with ground truth of "Severe Damage" as shown in Figure 8

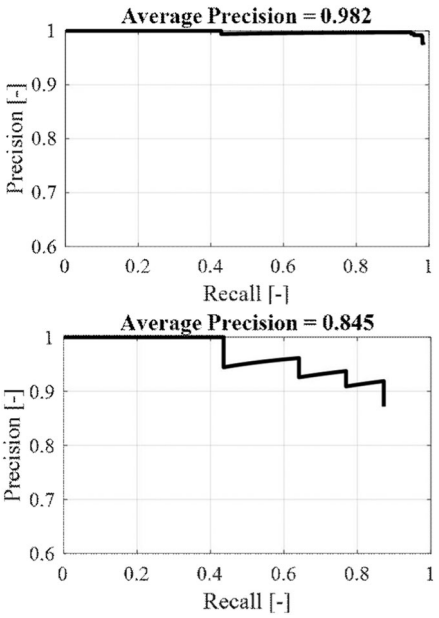


FIGURE 17 Recall–precision curve of training (upper) and testing (lower)

both high recall and precision. A detector with high recall but low precision retrieves many results, but most of its predicted labels are incorrect (e.g., incorrect bounding box locations, low IoU). A detector with high precision but low recall can localize the object very accurately once the object is successfully recalled, but only a very few results can be recalled. The average precision (AP) is often used to quantify the performance of an object detector (Girshick, 2015; Ren et al., 2017), which is determined as the area under the precision–recall curve. Mean AP (mAP) is defined as the mean of calculated APs for all classes. Figure 17 presents the precision–recall curve for training and testing. The mAP for both training and testing has demonstrated the applicability of YOLOv2 for detecting steel bars. The testing results in Figure 17 (lower) indicate there still exists room for improving the mAP with the larger training set, particularly improving the recall. It should be noted that the detection of steel bars is more difficult than detecting objects with a more regular pattern because the steel bars may buckle or fracture in a very complex way in different situations. Figure 18 provides sample detection images where the steel bars are localized by rectangular bounding boxes. Figure 18b (upper) shows the images that were wrongly classified by the traditional classification method, whereas Figure 18b (lower) shows the same images in which the exposed steel bars are localized by YOLOv2.

3.2.3 | DS determination

As illustrated in Figure 3, the proposed evaluation scheme integrates the classification model and object detection model to reinforce the identification of the most severe DS. Figure 19 presents the identification summary of a single classification model and the dual CNNs model using the confusion matrix method. Overall, the employment of the YOLOv2 object detector improves the identification accuracy of DS 3 by 7.5%. It should be noted that the classification accuracy can be potentially improved by expanding the training on the large data set. However, the training data for civil engineering applications is relatively limited as aforementioned. The YOLOv2 network demonstrated in this study can be considered as a local feature-oriented detector on the basis of existing

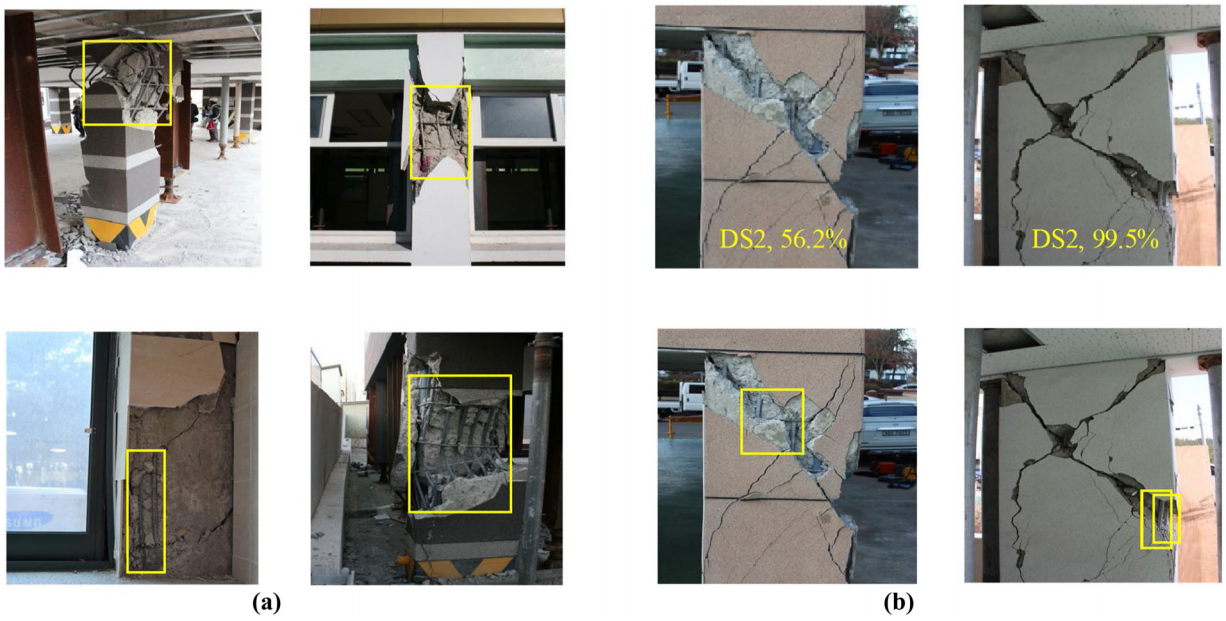


FIGURE 18 Detection of steel bars highlighted by yellow rectangular bounding boxes: (a) sample testing images, (b) detection of exposed steel bars (lower) in testing images wrongly predicted by classification model (upper)

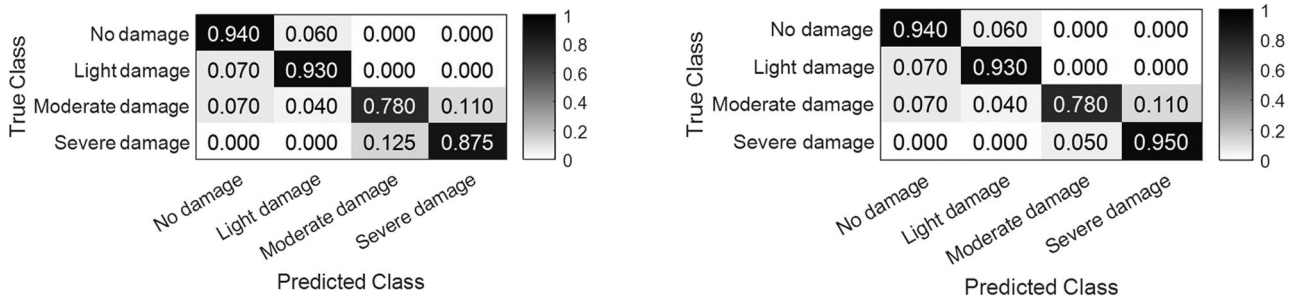


FIGURE 19 Confusion matrix with consideration of only classification model (left); both classification and object detection model (right)



FIGURE 20 Case study of an RC building with sample results (a) system-level identification: noncollapse and (b) component-level damage evaluation: severe damage with detection of steel exposure (left), and moderate damage (right)

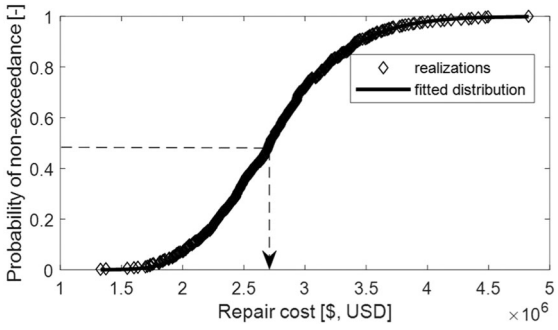


FIGURE 21 Repair cost distribution corresponding to the hypothetical case

classification networks to specially focus on the identification of DS 3 images when the training data set is relatively small.

3.2.4 | Repair cost evaluation

The DS determined by the proposed dual CNN algorithms can be integrated with the cost evaluation framework as presented in Section 2.5. For the purpose of illustration, a prototype RC building, as shown in Figure 20, is selected (Sim et al., 2015) to evaluate its damage and repair cost after an earthquake. In this study, the repair cost evaluation is based on the DSs of the available images collected. In summary, the dual CNNs algorithms determine that 17 of such columns are in DS 1, 26 in DS 2, and 14 in DS 3. Using the unit cost information provided in Table 3, the total repair cost of these columns is calculated. It should be noted that only the RC columns are considered in this study but similar procedures can be applied to all components. The process is repeated 10,000 times with Monte Carlo procedures to simulate the dispersion of the repair costs. Finally, the results are presented in a cumulative distribution function as shown in Figure 21. The cost simulation results can provide critical risk data for decision making and resource allocation during postdisaster reconstruction. For example, the decision maker can use the 50% probability of nonexceedance to identify the median repair cost for the building. In the example presented in Figure 21, the median repair cost is US$2.69 million for the prototype building.

4 | CONCLUSIONS

Rapid postdisaster damage estimation and cost evaluation of RC structures are becoming a crucial need for building owners and decision makers for risk management and resource allocation. This article proposed a rapid cost evaluation framework, which incorporates the state-of-the-art IPTs to quantify the structural damages and the integration with financial loss estimation of RC structures. Multiple innovations are presented in this article: (1) Both system-level and component-level classification models were trained successfully, which follows postdisaster DS quantification guidelines of RC structures. (2) A state-of-the-art real-time object detector, named YOLOv2, built on ResNet-50 was introduced and first implemented to demonstrate its applicability for detecting exposure of steel bars with 98.2% and 84.5% mAP in both training and testing process. (3) In comparison to a single classification

network, the implementation of the trained YOLOv2 combined with classification network improves the accuracy by 7.5% in identifying the most severe DS, which imposes critical threats for life safety and contributes the most to repair cost. (4) A novel integration of performance assessment framework with the damage detection methods is proposed and implemented to facilitate the repair cost evaluation, which can be easily conveyed to decision makers and stakeholders who lack engineering knowledge. Overall, the proposed framework shows the rapid inspection of the RC buildings with components is possible using image-based classification and detection techniques, although there still remains room to improve the recall in steel bars detection. The concept of the dual CNN scheme and the integration with cost estimation can be considered and extrapolated by other researchers for damage detection and loss evaluation of other structural types such as masonry, steel, and timber structures.

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